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RESEARCH ARTICLE

Effects of Different Noise Reduction Deep Learning Strategies on Computed Tomography Images

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ABSTRACT Noise reduction in Computed Tomography (CT) is very important for two reasons. On one hand, it can improve the quality of reconstructions, and on the other hand, significant dose reduction can be achieved, which is important due to the radiation risk to the patient. In this area, deep learning-based noise reduction possibilities are in the focus of interest. There are numerous different network models with various training strategies. In this paper, we examined different supervised training strategies with four different deep-learning architectures, and for the training, we only used mathematical phantoms with artificially simulated noise. The noise model was designed for a real multi-slice cone-beam CT and was performed in projection space on forward-projected mathematical phantoms. We validated the noise model and evaluated the effectiveness of different deep learning-based noise reduction models. We redesigned some of the deep learning models and analyzed their noise reduction effect using the noise power spectrum. We examined the noise reduction performance of the models on both real multi-slice cone beam CT scans and mathematical phantoms, with particular attention to their practical applicability.

INDEX TERMS Computed tomography, deep learning, noise reduction.

I. INTRODUCTION

In practice, various techniques are used in Computed Tomography (CT) imaging to improve the quality of reconstructions. Methods such as noise reduction, metal or streak artifacts reduction, beam hardening, and air correction helps reduce or remove artifacts that appear in images. The image reconstructed from the projections is adapted to the desired protocol using different imaging chains. Noise reduction is applied at multiple levels within the imaging chain. It can be performed in the projection and image domain, but different reconstruction methods also allow for noise reduction during the reconstruction process. In the case of Filtered Back Projection (FBP), appropriate kernels are used, while in iterative reconstruction methods, regularization steps are applied [1], [2], [3].

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It is advisable to perform noise reduction in multiple targeted steps, as different algorithms produce different noise patterns. Of course, noise reduction can also be associated with dose reduction, as in low-dose CT (LDCT), noisy low-dose images can be enhanced through noise reduction to a quality comparable to those reconstructed from a normal-dose projection series [4], [5], [6], [7], [8], [9], [10], [11], [12]. If noise filtering does not significantly alter the original noise texture or noise structure of the image, additional noise filtering can be applied [13]. For these reasons, it is essential not only to develop different noise reduction methods but also to understand their interactions, which requires a deeper noise analysis of the algorithms.

Several CT or CT-related AI-based algorithms used in industrial areas have been reported to preserve noise texture and noise power spectrum (NPS) characteristics. For example, GE Healthcare's TrueFidelity deep learning-based CT image reconstruction technology [14]. Philips (see in the white paper [15]) PRECISE AI deep learning-based image

reconstruction algorithm also preserves NPS characteristics. The ClariCT.AI algorithm by ClariPi [4], used in the CT field, also maintains the important NPS characteristics [13].

In this paper, we analyzed the denoising performance of four different deep learning-based noise reduction algorithms using two different training strategies, with particular attention to their practical applicability. We trained the models using only mathematical phantoms [16] and examined how much they changed the important NPS characteristics. The models trained on mathematical phantoms were tested on both mathematical phantom and real multi-slice cone-beam CT reconstruction. The objective of this approach was to assess the suitability of mathematical phantoms for training deep learning models designed for noise reduction. Noise reduction algorithms generally result in decreased resolution. This resolution loss can be reduced by weighted averaging of the noisy image and the noise-reduced image in the appropriate proportion, which is frequently used in practice. For this reason, we examined the noise spectra of the average of both noisy and denoised images and showed that it is not necessary to specifically design the models to preserve the characteristics of the noise.

II. MATERIALS AND METHODS

A. NOISE POWER SPECTRUM

The noise power spectrum (NPS) is a fundamental metric in medical imaging used to describe the spatial frequency characteristics of noise in a reconstructed image. It provides a quantitative measure of how noise is distributed across different spatial frequencies.

The NPS at spatial frequencies (f_x, f_y) is defined by

$$\text{NPS}(f_x, f_y) = \frac{1}{A} |\mathcal{F}_{2D}\{I(x, y)\}|^2, \quad (1)$$

where A is the total area, $I(x, y)$ represents the noise image and $\mathcal{F}_{2D}$ is the 2D Fourier Transform. A radial average of the NPS is computed as

$$\text{NPS}(f) = \frac{1}{N_f} \sum_{i=1}^{N_f} \text{NPS}(f_x, f_y) \quad (2)$$

where $f = \sqrt{f_x^2 + f_y^2}$, and N_f is the number of frequency components in each radial bin.

B. QUANTITATIVE INDICES

In this section we introduce the quantitative indices used in the paper to measure the accuracy of the reconstructions. There are many attempts to develop AI-based image quality assessment (IQA), but unfortunately these methods are still immature [17].

The Structural Similarity Index Measure (SSIM) is a metric used to assess the perceived quality of digital images. SSIM measures the similarity of two images, which provides useful information if a reference image is available. Research by Mason et al. [18] investigated how well different image quality metrics align with how radiologists evaluate the

diagnostic quality of MRI images. Their findings showed that SSIM performed worse than other quality metrics in terms of correlation with radiologists' assessments. In a separate study, Maruyama and colleagues [19] investigated how the size of regions of interest (ROI) affects SSIM calculations in X-ray imaging. Their research revealed that using larger ROIs tends to yield SSIM values closer to perfect similarity. They emphasized the importance of careful ROI selection and parameter configuration. To address these limitations, they suggested using a multi-scale SSIM (MS-SSIM) approach that considers different ROI sizes. Therefore, in this paper we used MS-SSIM [20] to compare a denoised image with the reference one.

Peak Signal-to-Noise Ratio (PSNR) is a widely used metric for measuring the quality of reconstructed CT images and compared to the standard (if the standard image is available). Let $x = (x_1, x_2, \dots, x_N)$ and $y = (y_1, y_2, \dots, y_N)$ be the processed and reference image, respectively. The PSNR index uses a logarithmic scale with units dB defined by the following formula:

$$\text{PSNR} = 20 \cdot \log_{10} \frac{M_I}{\sqrt{\text{MSE}}},$$

where

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N [x(i) - y(i)]^2,$$

and M_I is the dynamic range of the reference image. M_I is $2^B - 1$, where the maximal possible pixel value of the image is presented in B bits.

We used the effective dose saving factor (DSF) to compare the noise reduction effect of the different deep learning models. The virtual tube current dose multiplication factor (VTCM) [13] is calculated by this formula

$$\text{VTCM} = \left(\frac{N_0}{N_r} \right)^2,$$

where N_0 and N_r are the original and the reduced noise, respectively. It is a part of the advanced dose-modulation techniques in CT imaging. These techniques are used to ensure a balance between optimal image quality and minimal radiation exposure. To measure the noise, we used the standard deviation measured in homogeneous areas. Using VTCM factor, the effective dose saving factor is defined by the formula

$$\text{DSF} = \left(\frac{\text{VTCM} - 1}{\text{VTCM}} \right) \cdot 100.$$

C. NOISE SIMULATION

Several noise generators and normal-dose to low-dose generators have been proposed in recent decades [21], [22], [23], [24], [25]. For an overview of the physical background of the CT noise model, we recommend these excellent papers [26], [27], [28], [29]. To measure the effectiveness of the noise reduction algorithms, we developed a noise generation algorithm applied in projection space. The model

is similar to the method described in these papers [22], [24], with the difference that we took into account that the air-scan depends on the detector diodes. Since the detector we used was not photon-counting, the proportionality factor applied in this paper [22] was used as the basis for the photon number estimation for each detector diode.

The intensity I measured at the detector diodes is affected by both quantum noise and electronic noise. The quantum noise arises from fluctuations in the detected photon count, which follows a Poisson distribution, while the electronic noise originates from the background noise of the electronic hardware, typically modeled as Gaussian. Based on this, the measured intensity at the diodes can be described by the following expression:

$$I = P(\lambda) + N(m, \sigma^2), \quad (3)$$

where $P(\lambda)$ denotes a Poisson-distributed random variable with expected value λ , representing the mean number of detected photons, while the normal distributed electronic noise $N(m, \sigma^2)$ is assumed to have zero mean, i.e., $m = 0$. Since the measured intensity I is proportional to the number of photons, it is reasonable to assume that $E(I) = A \cdot \lambda$, where A is the conversion factor and $E(I)$ denotes the expected value of the intensity. Given that the variance of the intensity is $Var(I) = A^2\lambda + \sigma^2$, it follows that

$$\frac{Var(I)}{E(I)} = \frac{A^2\lambda + \sigma^2}{A\lambda} \approx \frac{A^2\lambda}{A\lambda} = A.$$

This implies that the conversion factor A can be estimated for each detector pixel based on a sufficient number of measurements, using the corresponding formula. Additionally, regarding the electronic noise, σ can also be estimated for each detector pixel with the aid of a sufficient number of air scans.

We performed the noise generation in projection space by forward projecting a phantom with a CT simulator. After that, the noisy projections were produced by generating noise on the noise-free projections using the above-mentioned conversion factor A and formula 3 and the projections were reconstructed using FBP.

In the section III-A, we describe the result of the simulated noise validation. An idealized homogeneous water phantom was used for validation. During the idealization process, the water phantom was scanned several times using AnyScan SPECT/CT (Mediso Ltd.), and then the reconstructed images were averaged and smoothed. In order to use the noise generator described above, the idealized water phantom was forward-projected using a CT simulator. We used the aforementioned AnyScan SPECT/CT geometry for the simulator. During the reconstructions, we also used the original AnyScan SPECT/CT multi-slice cone beam CT geometry, modeling the detector modules, the gaps between the modules, the rows and columns and size of the detector diodes, additional technical details are available on the official website. We did not use any image modification

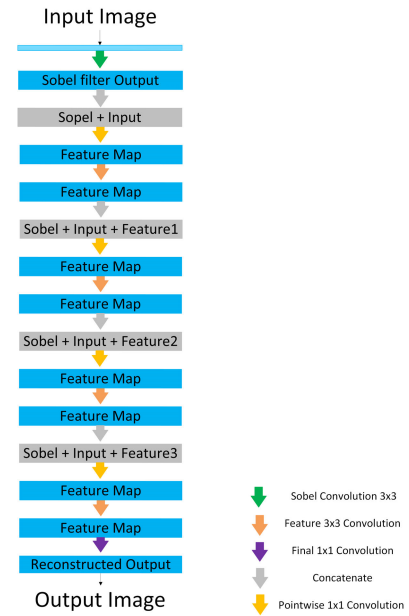


FIGURE 1. The structure of the used EDCNN network.

procedures, however in practice, CT imaging is the result of complex and composite image chains.

D. NEURAL NETWORKS AND TRAINING STRATEGIES

We selected four neural networks for the tests, which were redesigned when necessary. Each model was trained using two strategies. In both cases, the input files were the noisy mathematical phantoms. In the first strategy (Noise to noise-free, N2N), the target output images were the noise-free mathematical phantoms. In the other strategy (Noise to direct noise, N2DN), the target output images were the pure noise images. These images were obtained by subtracting the noise-free phantom reconstructions from those that were noisy with our noise generator. During training, we used min-max normalization.

Now we describe the four network architectures. The first is the Edge enhancement-based Densely Connected Neural Network (EDCNN) which was designed for noise reduction of low-dose CT images in this paper [30]. EDCNN effectively preserves image details while minimizing noise. The main process is to preprocess the image using an edge enhancement module that uses a trainable Sobel convolution to adaptively enhance edges of different intensities in the input images. This edge information is integrated into a dense network (DenseNet), which combines the original image and the processed data to improve the final result. Fig. 1 shows the diagram of the EDCNN.

Diffusion noise reduction models [31] learn to remove noise from data by first adding noise to clean samples and then training a neural network to reverse this process. During inference, they iteratively denoise a noisy input, gradually reconstructing a cleaner version. This approach is highly

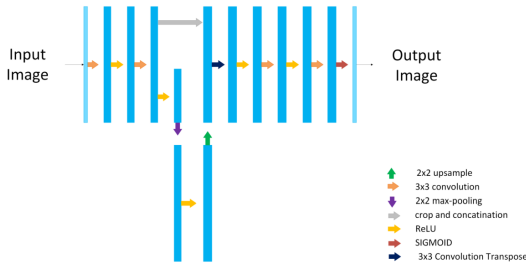


FIGURE 2. The structure of the used Diffusion architecture.

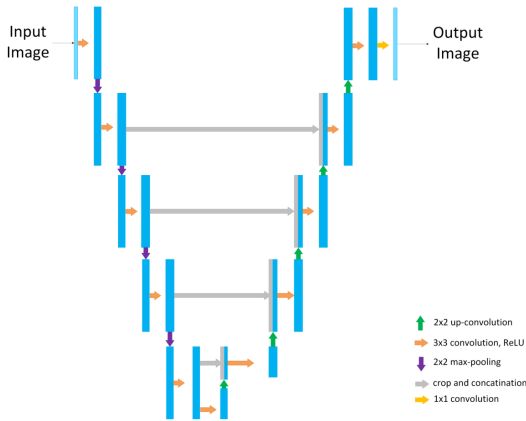


FIGURE 3. The structure of the used U-Net.

effective for image, audio, and medical data restoration, preserving details better than traditional noise reduction methods. We adapted the U-Net architecture used in the decoder part of the diffusion models to CT images, the diagram of the redesigned structure is shown in Fig. 2.

In practice, many U-Net models are used for CT image denoising, the reader is referred to these papers [4], [7], [32]. The diagram of the U-Net model we implemented and used can be seen in Fig. 3.

A generative adversarial network (WGAN) that uses the Wasserstein distance to estimate the difference between the distributions of denoised and normal-dose CT images can be found in this paper [5]. The used loss function is a feature-based loss calculated from a pre-trained VGG-19 network. The model encourages the generator to produce more realistic outputs. The discriminator assesses the realism of the generated images, ensuring quality improvement. Fig. 4 shows the diagram of the WGAN.

Obtaining real patient data for training is usually a difficult task, so we preferred to use only mathematical phantoms, which make it easier to obtain the data necessary for training. Therefore, in this study, we used realistic mathematical phantoms to train noise reduction algorithms, examining their usability in the training process. 4D XCAT phantoms [16] generated at 83 keV were utilized for the production of mathematical phantoms.

Table 1 and 2 present the training parameters and relative execution times. The dataset was partitioned such that approximately 80% was allocated for training, 10%

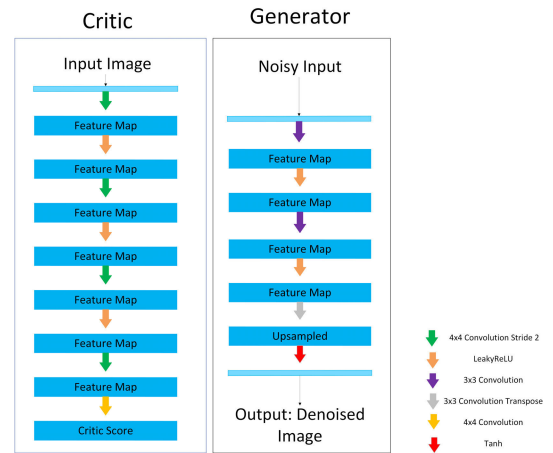


FIGURE 4. The structure of the used WGAN.

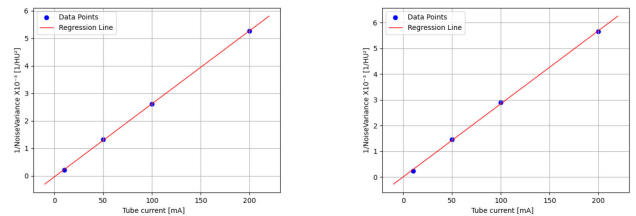


FIGURE 5. Regression lines. Arranged from left to right: original measurements based on water phantom, measurements on simulated data.

for validation, and 10% for testing. As all models were trained under identical conditions, their training durations are reported relative to the training time of the EDCNN model. Inference time measures how long it takes a trained model to process and denoise a single medical slice (512×512), relative to a unit EDCNN model. The time required for image averaging was also reported relative to the per-slice inference time of the EDCNN model, which was used as the reference unit.

Both the training of the deep learning models and the execution of the CT reconstruction algorithms were performed using GPUs. The use of GPUs has become increasingly important for accelerating the computationally intensive processes involved in complex reconstructions. We recommend the following papers for readers interested in this topic [33], [34], [35], [36].

III. RESULTS

A. VALIDATION OF SIMULATED NOISE

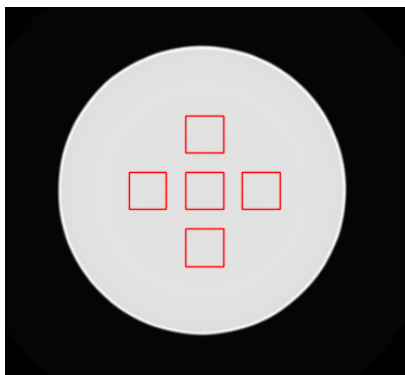
It is well known that the reciprocal of the tube current and the noise variance are proportional (see for example [37]). In the first step, we examine the noise model from this perspective. We used the water phantom described in section II-C, on which we measured the variance associated with different current levels on both real and noise-simulated reconstructions (Fig. 9). The real acquisitions at four tube current levels (10, 50, 100 and 200 mAs) were obtained

TABLE 1. Training parameters and relative execution times of the N2N strategy.

Model	WGAN	EDCNN	Diffusion	U-Net
Epochs	100	100	100	100
Best performance at	epoch 68	epoch 97	epoch 100	epoch 51
Best validation loss:	0.000012	0.000008	0.000013	0.000025
Loss Function	MSE	MSE	MSE	MSE
Optimizer	Adam with weight decay	Adam with weight decay	Adam with weight decay	Adam with weight decay
Learning Rate	0.0001	0.0001	0.0001	0.0001
Batch Size	4	4	4	4
Patience	5	5	5	5
Weight Decay	0.00001	0.00001	0.00001	0.00001
Training time	0.07	1	0.6	0.67
Interference time per slice	1.22	1	0.09	0.04
Averaging time per slice	0.087	0.087	0.087	0.087

TABLE 2. Training parameters and relative execution times of the N2DN strategy.

Model	WGAN	EDCNN	Diffusion	U-Net
Epochs	100	100	100	100
Best performance at	epoch 20	epoch 78	epoch 3	epoch 38
Best validation loss:	0.000276	0.000287	0.000077	0.000293
Loss Function	MSE	MSE	MSE	MSE
Optimizer	Adam with weight decay	Adam with weight decay	Adam with weight decay	Adam with weight decay
Learning Rate	0.0001	0.0001	0.0001	0.0001
Batch Size	4	4	4	4
Patience	5	5	5	5
Weight Decay	0.00001	0.00001	0.00001	0.00001
Training time	0.038	1	0.026	0.974
Interference time per slice	0.6	1	0.09	1.48
Averaging time per slice	0.080	0.080	0.080	0.080

**FIGURE 6.** Smoothed water phantom and the selected ROIs.

using AnyScan SPECT/CT. The corresponding simulated projections were generated using the method described in section II-C. This relationship is shown in Fig. 5 using a regression line. The slopes of the regression lines are also nearly identical, 0.02655 for the real case and 0.02836 for the simulated noise.

In the next step of the validation, we estimated the noise using standard deviation measured in five areas of equal size (see Fig. 6). The results of the measurements are shown in Table 3. The percentage error of the simulated noise at tube current levels of 10, 50, 100 and 200 mAs are 5.9, 2.2, 5.4 and 2.3, respectively, which means an average error of 3.9%.

We also validated the noise model using NPS. The NPS was calculated and averaged on the ROIs presented in Fig. 6. Fig 7 shows the noise power spectra, which confirm that the

noise in the simulated images shows a similar structure as in the real images. The noise distribution is also nearly identical, and the noise peaks are located in nearly the same frequency range, with a slightly difference only in the amplitude of the noise.

The statistical validation of the simulated noise was performed using both the Kolmogorov–Smirnov test and the Cramér–von Mises test. Both are designed to assess the agreement between two distributions, though their sensitivity may vary depending on the sample characteristics. Sampling was based on FBP reconstructions of measured and simulated water phantom acquired at various tube current levels, using the five ROIs shown in Fig. 6 as sampling regions. Each ROI provided 400 sample points. To mitigate distortions introduced by the FBP reconstruction, the sample means were normalized to zero.

The results of the tests are presented in Table 4. At all reasonable significance levels, the null hypothesis is not rejected, indicating that no statistically significant difference was detected between the distributions. The noise model performed worst at a tube current of 100 mAs, yet even at 10 mAs it remained statistically consistent.

B. NOISE ANALYSIS

Table 5 compares the noise reduction effect of the different models. Standard deviations were computed on a water phantom on the ROI shown in Fig. 9.

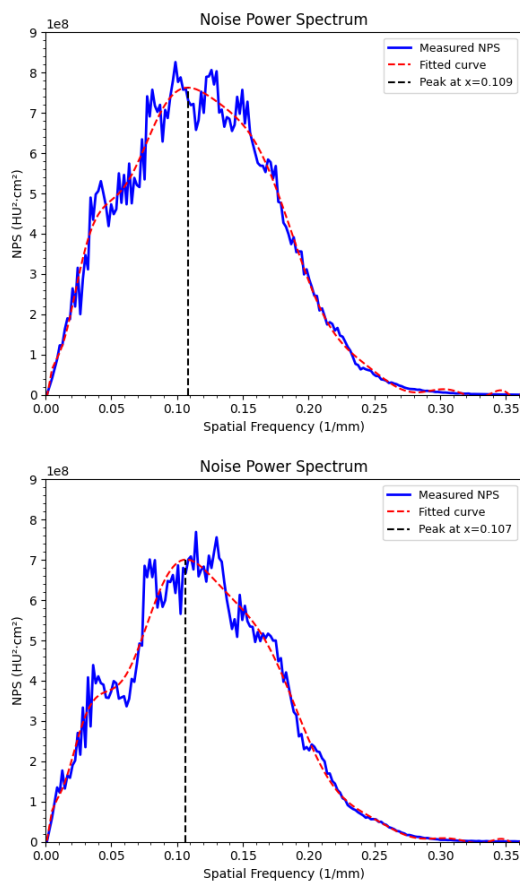
The statistical data presented in Table 5 demonstrate that the models achieved a minimum average noise reduction

TABLE 3. Measured and simulated noise level at each ROI and their mean in HU.

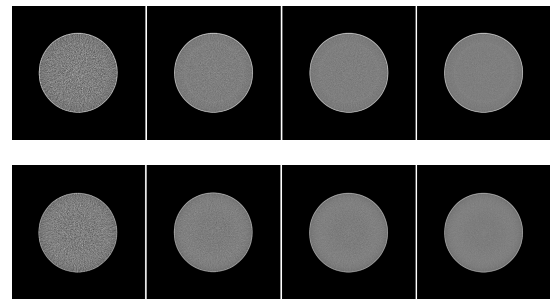
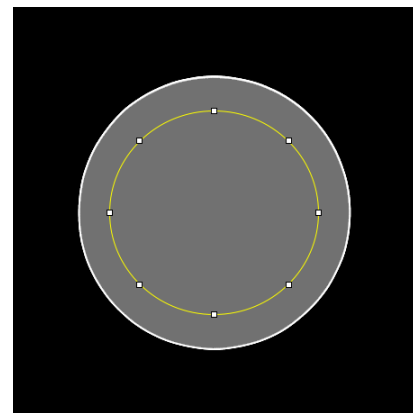
ROI	10 mAs		50 mAs		100 mAs		200 mAs	
	Measured	Simulated	Measured	Simulated	Measured	Simulated	Measured	Simulated
1	65.71	67.45	27.46	27.78	20.13	19.53	13.91	13.89
2	72.31	70.61	30.03	29.03	21.24	19.54	14.9	14.25
3	72.64	64.99	27.81	27.51	19.69	18.45	13.82	14.01
4	74.48	66.83	26.98	26.95	20.85	19.68	14.52	14.05
5	73.37	67.58	28.44	26.39	19.22	18.46	13.9	13.2
Mean	71.702	67.492	28.144	27.532	20.226	19.132	14.21	13.88

TABLE 4. The p-values obtained from the two-sample Kolmogorov–Smirnov and Cramér–von Mises tests were averaged over the ROIs shown in Fig. 6.

Tube current	10 mAs	50 mAs	100 mAs	200 mAs
Kolmogorov–Smirnov test	0.5481 ± 0.3566	0.6630 ± 0.2897	0.2704 ± 0.1229	0.6041 ± 0.2928
Cramér–von Mises test	0.5032 ± 0.3713	0.6132 ± 0.3155	0.2745 ± 0.1284	0.6409 ± 0.3154

**FIGURE 7.** Noise Power Spectrum on 10 mAs images. Original and synthesized noise measured on water phantom.

effect of 80.72% for the N2N and 24.21% for the N2DN strategies. In the case of the N2DN strategy, the noise reduction effect is practically independent of the tube current, while the N2N strategy is slightly more sensitive to it. The average noise reduction effect when choosing the N2N strategy was greater for all models than for the N2DN strategy, except for the U-Net model. The EDCNN and WGAN structures using the N2N strategy were slightly effective than using the N2DN strategy. However, in the case

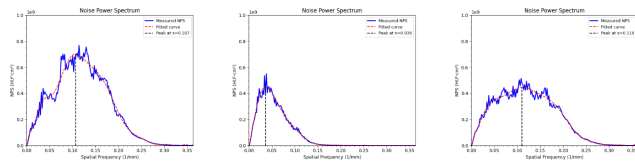
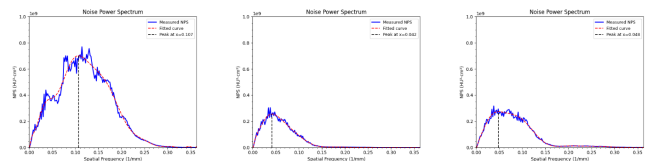
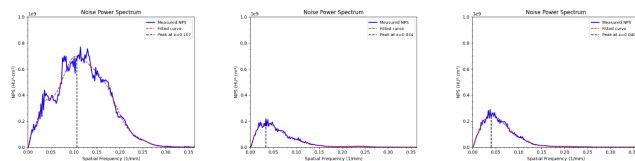
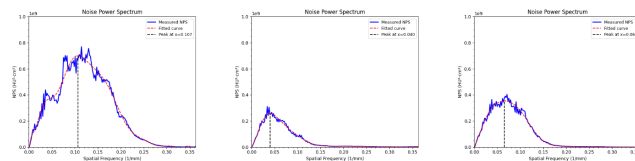
**FIGURE 8.** The first row shows the real and the second row shows the simulated noisy water phantom. From left to right, the tube current used is 10, 50, 100, 200 mAs. Display window/level=400/0 HU.**FIGURE 9.** The water phantom and the ROI used for noise measurement.

of the Diffusion model, the noise reduction effect of the N2N strategy is more than three times that of the N2DN strategy.

Of course, a lower standard deviation does not necessarily mean better image quality. Therefore in the following sections, we evaluate image quality on realistic human mathematical phantoms and real reconstructions. In this section, we continue the analysis by evaluating the noise power spectra measured on the water phantom, which provide information about the frequencies at which the noise energy is concentrated. For the different models, Fig. 10, 11, 12 and 13 show the noise structure on the noise-simulated scans using the N2N and N2DN strategies.

TABLE 5. The Simulated row shows the standard deviation in HU, the other rows show the effective dose saving factors of the different models compared to the simulated noise for the N2N and N2DN training strategies.

Strategy	10 mAs		50 mAs		100 mAs		200 mAs		Average	
	N2N	N2DN	N2N	N2DN	N2N	N2DN	N2N	N2DN	N2N	N2DN
Simulated	64.65 (HU)		26.19 (HU)		18.39 (HU)		13.29 (HU)		30.63 (HU)	
Diffusion	86.05	25.64	81.94	24.61	78.89	22.78	76.01	23.82	80.72	24.21
EDCNN	93.08	91.11	96.61	92.15	96.38	91.35	95.27	89.97	95.34	91.15
U-net	90.2	96.4	84.71	85.1	81.69	84.88	77.74	84.45	83.59	87.71
WGAN	91.04	81.92	89.88	81.05	88.71	79.87	87.6	79	89.31	80.46

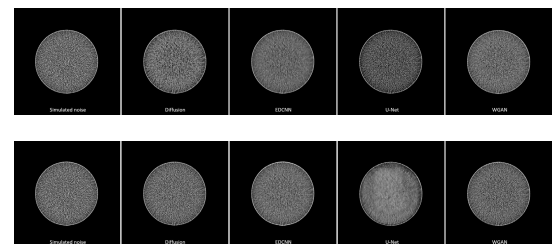
**FIGURE 10.** NPS of the Diffusion model on 10 mAs reconstructions. Arranged from left to right: simulated noisy image, Diffusion with N2N, Diffusion with N2DN.**FIGURE 13.** NPS of the WGAN model on 10 mAs reconstructions. Arranged from left to right: simulated noisy image, WGAN with N2N, WGAN with N2DN.**FIGURE 11.** NPS of the EDCNN model on 10 mAs reconstructions. Arranged from left to right: simulated noisy image, EDCNN with N2N, EDCNN with N2DN.**FIGURE 12.** NPS of the U-Net model on 10 mAs reconstructions. Arranged from left to right: simulated noisy image, U-Net with N2N, U-Net with N2DN.

In general, the N2N strategy provides stronger noise reduction, shifting the noise peak towards lower frequencies. The only exception to this was the Diffusion model with N2DN strategy. The Diffusion model reduces the noise less using the N2DN strategy, and as a consequence it preserves its characteristics better and changes the NPS peak only slightly (see Fig. 10). The EDCNN, U-Net and WGAN models were not very sensitive to the training strategies, however, based on the noise spectra (see Fig. 11, 12, 13), the N2DN strategy better preserves the noise structure in the case of U-Net and WGAN.

Fig. 14 presents the noisy water phantom using different models.

C. NOISE ANALYSIS ON AVERAGED RECONSTRUCTIONS

Traditional noise reduction algorithms usually do not preserve the characteristic of the noise spectrum. Therefore, in practice, the noisy and the denoised images are

**FIGURE 14.** The first row shows the simulated noise, Diffusion, EDCNN, U-Net, WGAN images with N2N. The second row shows the simulated noise, Diffusion, EDCNN, U-Net, WGAN images with N2DN. The tube current was 10 mAs. Display window/level=200/0 HU.

weighted-averaged, thereby preserving the more important structures, including the noise structures and NPS peaks. In this section, we examined the noise power spectrum when the noisy and denoised reconstructions were averaged. Of course, in such a case, the noise reduction effect will be smaller, but this can be compensated by applying the noise reduction algorithms multiple times, iteratively.

If a model is trained with noisy and noise-free image pairs, it will usually oversmooth the reconstructed image and may remove diagnostically important details. Section III-B shows that the peak of the noise spectrum was only preserved by the Diffusion model when the model was trained with the N2DN strategy. However, according to Table 5, the noise reduction effect is the smallest in this case.

Based on Figures 15-18, It can be seen that with this averaging method, the noise spectrum of almost all models retained the noise structure of the original image, including the NPS peak, except for the Diffusion model that was trained with the N2N strategy. It should be noted that with this model the noise reduction was so effective that the peak cannot be clearly identified. Fig. 19 shows the averaged water phantom reconstructions using different models and strategies.

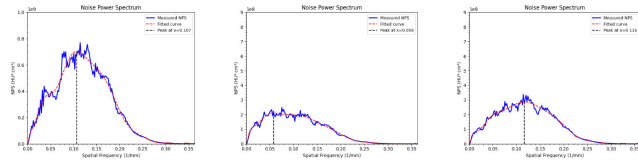


FIGURE 15. NPS of the Diffusion model on 10 mAs reconstructions. Arranged from left to right: simulated noisy image, the average over Diffusion with N2N and the simulated noisy image, the average over Diffusion with N2DN and the simulated noisy image.

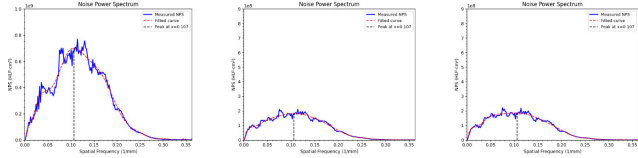


FIGURE 16. NPS of the EDCNN model on 10 mAs reconstructions. Arranged from left to right: simulated noisy image, the average over EDCNN with N2N and the simulated noisy image, the average over EDCNN with N2DN and the simulated noisy image.

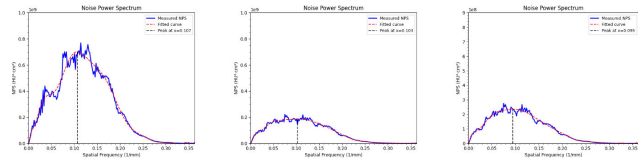


FIGURE 17. NPS of the U-Net model on 10 mAs reconstructions. Arranged from left to right: simulated noisy image, the average over U-Net with N2N and the simulated noisy image, the average over U-Net with N2DN and the simulated noisy image.

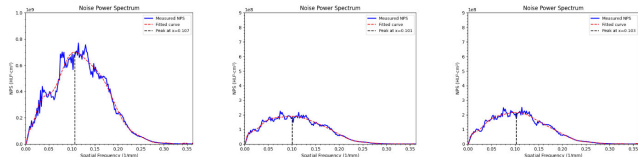


FIGURE 18. NPS of the WGAN model on 10 mAs reconstructions. Arranged from left to right: simulated noisy image, the average over WGAN with N2N and the simulated noisy image, the average over WGAN with N2DN and the simulated noisy image.

D. MATHEMATICAL PHANTOM CASE

Since the degree of noise reduction does not equal better image quality, in this section we examine the effect of each model on a mathematical phantom. For this purpose, we used the XCAT phantom that is a realistically simulated human body phantom.

We evaluated the reconstructions with noise added at 25 mAs. The noise reduction was performed in projection space as described in Section II-C, and then the cone beam scans were reconstructed using FBP. Fig. 20 presents the phantom used with the ROI for standard deviation, as well as the noise-added phantom and the effect of the investigated noise-reduction AI models on the noise-added phantom.

Table 6 shows the effective dose saving factors, PSNR and MS-SSIM indices measured on the phantom compared

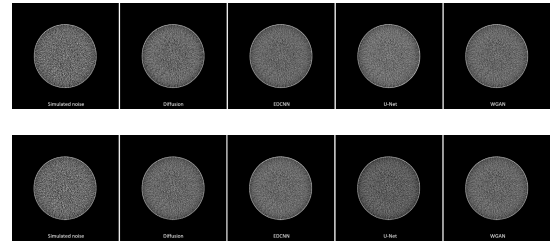


FIGURE 19. The first row shows the simulated noise, Diffusion, U-Net, WGAN images with N2N related to the average. The second row shows the simulated noise, Diffusion, EDCNN, U-Net, WGAN images with N2DN related to the average. The tube current was 10 mAs. Display window/level=200/0 HU.

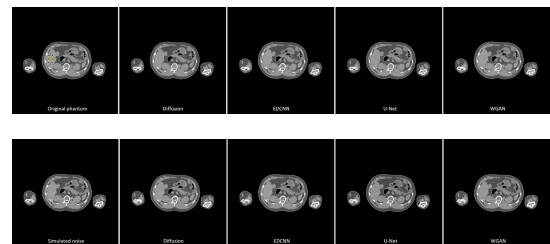


FIGURE 20. The first row shows the original phantom with the ROI, Diffusion, EDCNN, U-Net, WGAN reconstructions with N2N. The second row shows the simulated noise, Diffusion, EDCNN, U-Net, WGAN reconstructions with N2DN. The tube current was equivalent to 25 mAs. Display window/level=500/0 HU.

to the reference phantom. The table contains the means and standard deviations of PSNR and MS-SSIM indices measured across four detail-rich regions of the phantom. The reference phantom is the leftmost image in the first row of Fig. 20.

Noise-induced streak artifacts are generally removed in the projection space [38], but these artifacts can also be removed in the image space using deep learning. Fig. 20 shows that all four deep learning structures are able to reduce streak artifacts, but not surprisingly, only when using the N2N strategy. With the N2DN strategy, each model is able to reduce noise without generating major artifacts. The greatest noise reduction can be achieved with EDCNN by using the N2N strategy. Based on the MS-SSIM indices, the image most similar to the reference was achieved by the Diffusion, EDCNN and U-Net networks using the N2N strategy. The WGAN network did not perform very well with N2N strategy, as shown in Table 6 and Fig. 20. Due to the streak artifacts, the N2DN strategy gave less good PSNR and MS-SSIM values, however, except for the EDCNN model, all of them retained the edges nicely.

E. PATIENT CASE

In this section, we examine the applicability of AI models to real CT scans. To evaluate real reconstructions, we chose a relatively artifact-free real scan to avoid differences caused by artifacts. The original reconstructed real projections is the leftmost image in the first row of Fig. 21, which is a real scan of an anonymized patient.

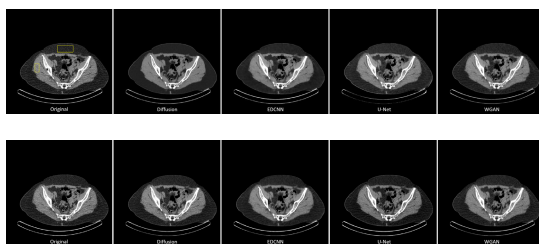
TABLE 6. The effective dose saving factors, PSNR and MS-SSIM index of the N2N and N2DN training strategies mathematical phantoms with 25 mAs equivalent noise.

Strategy	SD		PSNR		MS-SSIM	
	N2N	N2DN	N2N	N2DN	N2N	N2DN
Simulated	41.54	(HU)	32.57 ± 0.34 dB	(HU)	0.9555 ± 0.0068	(HU)
Diffusion	97.98	45.35	39.40 ± 0.35	34.14 ± 0.29	0.9905 ± 0.0013	0.9708 ± 0.0031
EDCNN	98.7	96.34	41.49 ± 0.43	39.28 ± 0.40	0.9928 ± 0.0014	0.9877 ± 0.0016
U-Net	97.91	90.8	40.79 ± 0.60	37.23 ± 0.15	0.9915 ± 0.0016	0.9865 ± 0.0018
WGAN	96.42	90.81	40.00 ± 0.35	37.64 ± 0.32	0.9894 ± 0.0015	0.9847 ± 0.0016

During the training of deep learning structures, we exclusively used artificially noised XCAT phantoms with different noise levels. Direct application of the models on real images can be very dangerous even if real images are used during training, therefore in practice, a weighted average of the denoised and original images is typically used. In this paper, we mixed the original and denoised images in a 50-50 percent ratio. Fig. 21 shows the original reconstructed image and the direct output of the models, while Fig. 22 shows the result of denoising with reconstructions mixed in 50-50 percent ratio.

We evaluated the degree of noise reduction in fat, muscle and bone regions and summarized it in Table 7 and 8. The first images in the first row of Fig. 22 and Fig. 21 also show the ROIs used for the measurement.

As seen in the leftmost image of the top row in Fig. 21, the chosen real reconstruction was relatively streak-free. This is important to note because, in the previous section, we observed that the N2N strategy was able to reduce these artifacts with all models, whereas the N2DN strategy was not. Similar to the measurements on the mathematical phantom, Table 7 confirms that using real scans, greater noise reduction can also be achieved with the N2N strategy. The difference between the strategies is most pronounced in the case of the Diffusion model, while the other deep learning models show approximately the same noise reduction effect for both strategies. The strongest noise reduction is achieved with the Diffusion and EDCNN models using N2N strategy. The effective dose saving factor of the Diffusion model is 97.23, while that of the EDCNN model is 94.98. The Diffusion model performed worst with N2DN strategy.

**FIGURE 21.** Original and noise reduced reconstructions with the selected ROIs. The first row shows the original, Diffusion, EDCNN, U-Net, WGAN images with N2N. The second row shows the original, Diffusion, EDCNN, U-Net, WGAN images with N2DN. The tube current was 100 mAs. Display window/level=400/50 HU.

The effective dose saving factor in averaged images in soft tissues is approximately between 58 and 70 for the N2N

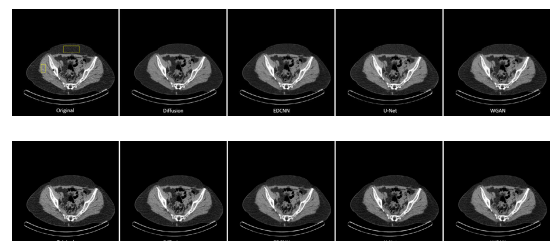
TABLE 7. The effective dose saving factors on denoised reconstructions with N2N and N2DN training strategies.

	Fat		Muscle		Bone	
	N2N	N2DN	N2N	N2DN	N2N	N2DN
Original	22.94	22.94	36.68	36.68	236.11	236.11
Diffusion	97.23	17.39	95.68	41.35	58.69	2.65
EDCNN	94.98	89.59	93.31	90.27	-8.72	11.63
U-Net	86.37	79.05	88.14	81.19	-1.81	-1.4
WGAN	84.85	71.44	91.23	83.96	52.62	-16.11

TABLE 8. The effective dose saving factors on weighted-averaged reconstructions with N2N and N2DN training strategies.

	Fat		Muscle		Bone	
	N2N	N2DN	N2N	N2DN	N2N	N2DN
Original	22.94	22.94	36.68	36.68	236.11	236.11
Diffusion	68.38	10.19	68	26.72	34.37	1.59
EDCNN	70.07	62.65	67.6	65.26	-3.62	8.98
U-Net	60.92	53.04	61.94	55.79	0.09	0.63
WGAN	58.55	47.32	65.55	57.65	29.56	-6.5

strategy and between 47 and 66 for the N2DN strategy, except the Diffusion model, that achieved only between 10 and 27. Fig. 21 shows that using the N2N strategy, the denoised images become heavily oversmoothed. This is most evident in the case of the EDCNN model. However, the Diffusion model preserves the edges with both strategies, even when using the N2N strategy. The EDCNN model modifies the details of the reconstructions the most. It should be noted that the preservation of reconstruction details can be achieved with the U-Net and WGAN models using the N2DN strategy. In these cases (see Fig. 22), no significant artifacts are visible in the averaged images.

**FIGURE 22.** Original and weighted-averaged reconstructions with the selected ROIs. The first row shows the original, Diffusion, EDCNN, U-Net, WGAN images with N2N. The second row shows the original, Diffusion, EDCNN, U-Net, WGAN images with N2DN. The tube current was 100 mAs. Display window/level=400/50 HU.

IV. DISCUSSION

In this paper, we compared four noise reduction deep learning models based on noise power spectrum, effective dose saving factor, PSNR and MS-SSMI index, on realistic mathematical phantoms and real reconstructions. Note that the original Diffusion and U-Net architecture has been slightly redesigned. We applied two different training strategies for each model and exclusively used artificially noisy mathematical phantoms during training. The evaluation focused on the noise structure and the preservation of details in the reconstructed images.

From the NPS, it can be seen that in order to preserve the noise structure, it is beneficial to consider the denoised and noisy images with weighted averaging. If preserving the structure of the noise is not absolutely necessary, the N2N strategy, using Diffusion, U-Net and WGAN networks, can significantly reduce the noise while preserving reconstruction details without the need to average the noisy and denoised reconstructions. Considering both the degree of noise reduction and the quality of real and phantom reconstructions, the redesigned Diffusion network proves to be the most optimal choice using the N2N strategy.

It should be noted, however, that when employing the N2N strategy, the results on real patient data exhibited substantial oversmoothing and artifacts. This observation supports the notion that the use of real acquisitions may be necessary for effectively training such models. In this study, the models were assessed solely in the abdominal region. Nevertheless, future work will focus on extending the evaluation to additional anatomical areas.

Finally, it is worth noting that deep learning models designed to preserve the characteristics of the noise power spectrum are not necessarily alternatives to conventional denoising algorithms. Rather, they can be complementary, and their combination may lead to further noise reduction (see, for example, in [13]).

V. CONCLUSION

We used a noise model designed and validated for a real CT device to simulate noise to mathematical phantoms, which were then used to train four different deep learning models with two different training strategies.

When using artificially noised mathematical phantoms during the training of deep learning networks, the images become over-smoothed. However, based on the NPS analysis for all four examined models, it can be stated that the noise structure of the reconstructed images can be preserved if we use a weighted average of the noisy and noise-free images. Without this, the structure of the noise is altered by the examined networks, with the only exception being a redesigned Diffusion model with the N2DN strategy. With this method, the details of the images can also be preserved, although the degree of noise reduction will be smaller. Considering only the degree of noise reduction and image quality of realistic mathematical phantom and real reconstruction, the N2N strategy using the Diffusion network

proves to be the most optimal choice. In this case, the effective dose saving factor in averaged images and in soft tissues is an average of 68.18. However, without using the weighted method, an average of 96.46 effective dose saving factor in soft tissues can be achieved in real reconstructions without significant artifacts.

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